Python Assignment for Week 1

Python Assignment 1: Fundamentals of Python

Course: Introduction to Python
Topics Covered:
Variables, Comments, Strings, Numbers, Operators, Casting, Data Types, Booleans

Instructions:

- Answer all questions.
- Write clean and readable code.
- Use comments to explain your logic where required.
- Do not use advanced concepts (no loops, no functions, no lists).

Section A: Variables & Comments (10 marks)

1. Create a variable called student_name and assign your full name to it.
2. Create a variable called age and assign your age.
3. Create a variable called is_student and assign a boolean value.
4. Add **comments** explaining what each variable represents.
5. Print all the variables.

Section B: Strings (10 marks)

1. Create two string variables:
    - first_name
    - last_name
2. Join them together to form a full name and store it in full_name.
3. Print the following sentence using the variable:
4. My name is <full_name>
5. Convert the full_name to:

- Uppercase
- Lowercase

Section C: Numbers & Operators (15 marks)

1. Create two variables:
    - num1 = 25
    - num2 = 7
2. Perform the following operations and print the results:
    - Addition
    - Subtraction
    - Multiplication
    - Division
    - Modulus
3. Add a comment explaining what each operator does.

Section D: Data Types (15 marks)

1. Create the following variables:
    - An integer
    - A float
    - A string
    - A boolean
2. Use the type() function to print the data type of each variable.
3. Write a comment explaining what a data type is.

Section E: Type Casting (15 marks)

1. Create a variable:
2. number_as_string = "50"

3. Convert it to:
    - An integer
    - A float
4. Add 10 to both converted values and print the results.
5. Explain with comments why casting is important.

Section F: Booleans & Comparisons (20 marks)

1. Create two variables:
    - x = 10
    - y = 20
2. Compare them using:
    - ==
    - !=
    - >
    - <
    - >=
    - <=
3. Print the result of each comparison.
4. Write one sentence explaining what a boolean value means.

Section G: Mini Practice Task (15 marks)

Write a small program that:

1. Stores a person's name and year of birth.
2. Calculates their age (use a fixed year like 2025).
3. Checks if the person is **18 years or older**.
4. Prints:
5. Name: <name>

6. Age: <age>
7. Adult: <True/False>

Submission Rules

- Save your file as:
  python_assignment_1.py
- Code must run without errors.
- Late submission attracts penalty. (deadline Monday 3:00am)

Marking Guide

- Correct syntax: 40%
- Proper use of variables & data types: 30%
- Comments & readability: 20%
- Output correctness: 10%